Project: Worked examples
Title: Weld group (elastic line method)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

A group of straight fillet welds in a plane under in-plane forces (with eccentricity, so torsion), an out-of-plane force and out-of-plane moments: the resultant force per unit length at every weld end, the critical point, the throat stress against the design strength and the leg size required.

WHAT IT CALCULATES	RELATION
Welds as lines of unit throat	$L_w = \Sigma\, l$, centroid $x_c = \Sigma\, l \cdot x_m / L_w$
Unit second moments	$I_{ux} = \Sigma\,[l(y_m - y_c)^2 + l \cdot \Delta y^2/12]$, I_{uy} likewise, product $I_{uxy} = \Sigma\,[l(x_m - x_c)(y_m - y_c) + l \cdot \Delta x \cdot \Delta y/12]$, $J_u = I_{ux} + I_{uy}$
Torsion about the centroid	$T = F_y(e_x - x_c) - F_x(e_y - y_c) + M_z$
At each weld end	direct F/L_w, torsional $T \cdot r/J_u$ (perpendicular to r), out-of-plane $F_z/L_w + [(M_x \cdot I_{uy} - M_y \cdot I_{uxy}) \cdot y' + (M_y \cdot I_{ux} - M_x \cdot I_{uxy}) \cdot x']/(I_{ux} \cdot I_{uy} - I_{uxy}^2)$ with F_z's eccentricity added to M_x and M_y (this reduces to $M_x \cdot y'/I_{ux} + M_y \cdot x'/I_{uy}$ only for a group symmetric about a centroidal axis); vector sum f_r
Throat and weld stress	$a = 0.707 \cdot s$, $\tau = f_{r,max}/a$; required leg $s_{req} = f_{r,max}/(0.707 \cdot f_w)$

Basis: Welds treated as lines: torsional and bending properties[1](11th ed., Ch. 9, Tables 9-1 and 9-2); Treating the weld as a line[2](1966, §7.4); Simplified method for design resistance of fillet welds (resistance input)[3](cl. 4.5.3.3).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
welds	Weld lines	3 row(s)		drawing: BRK-010 rev A	any
s	Fillet leg	8	mm	drawing: BRK-010 rev A	3 … 25 mm
f_w	Design strength of the throat	220	MPa	assumption: S275, EN 1993-1-8 simplified method: 410/√3/(0.85 × 1.25) ≈ 223 MPa, rounded down.	50 … 600 MPa
F_x	In-plane force, x	0	kN	assumption: No horizontal load.	-100000 … 100000 kN
F_y	In-plane force, y	-30	kN	spec: Hoist rating with dynamic factor	-100000 … 100000 kN
F_z	Out-of-plane force (tension positive)	0	kN	assumption: No out-of-plane load.	-100000 … 100000 kN

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
e_x	Point of application, x	250	mm	drawing: BRK-010 rev A	-100000 … 100000 mm
e_y	Point of application, y	100	mm	drawing: BRK-010 rev A	-100000 … 100000 mm
M_x	Out-of-plane moment about x (tension on +y side)	0	kN·m	assumption: None.	-100000 … 100000 kN·m
M_y	Out-of-plane moment about y (tension on +x side)	0	kN·m	assumption: None.	-100000 … 100000 kN·m
M_z	Additional in-plane torque	0	kN·m	assumption: None.	-100000 … 100000 kN·m

TABLE ROW	VALUES
welds[1]	x1 = 0 mm; y1 = 0 mm; x2 = 80 mm; y2 = 0 mm
welds[2]	x1 = 0 mm; y1 = 0 mm; x2 = 0 mm; y2 = 200 mm
welds[3]	x1 = 0 mm; y1 = 200 mm; x2 = 80 mm; y2 = 200 mm

3. Assumptions

A1. Elastic, static; the connected parts are rigid and the weld stress varies linearly with distance from the centroid. (Conservative against the instantaneous-centre method.)
A2. Fillet welds only, all of one leg size; the throat is 0.707·s. (Equal-leg fillet.)
A3. The design strength of the throat comes from S4 (EN 1993-1-8 simplified method) or a user allowable with its source. (No resistance table ships.)
A4. f_w (Design strength of the throat) is taken as 220 MPa. (S275, EN 1993-1-8 simplified method: 410/√3/(0.85 × 1.25) ≈ 223 MPa, rounded down.)
A5. F_x (In-plane force, x) is taken as 0 kN. (No horizontal load.)
A6. F_z (Out-of-plane force (tension positive)) is taken as 0 kN. (No out-of-plane load.)
A7. M_x (Out-of-plane moment about x (tension on +y side)) is taken as 0 kN·m. (None.)
A8. M_y (Out-of-plane moment about y (tension on +x side)) is taken as 0 kN·m. (None.)
A9. M_z (Additional in-plane torque) is taken as 0 kN·m. (None.)

4. Working

[11] $L_w = \Sigma\ l$
$= 80 + 200 + 80$
$= 360\ \text{mm}$
Total weld length

[12] $x_c = \Sigma\ l \cdot x_m\ /\ L_w$
$= 17.778\ \text{mm}$
$= 17.778\ \text{mm}$
Centroid x

[13] $y_c = \Sigma\ l \cdot y_m\ /\ L_w$
$= 100\ \text{mm}$
$= 100\ \text{mm}$
Centroid y

[14] $I_{ux} = \Sigma\ [l(y_m - y_c)^2 + l \cdot \Delta y^2/12]$ [1](Table 9-2)
$=$ about the centroid
$= 2266700\ \text{mm}^3$
Unit second moment about x

[15] $I_{uy} = \Sigma\ [1(x_m - x_c)^2 + 1 \cdot \Delta x^2/12]$ [1](Table 9-2)
= about the centroid
= 227560 mm³
Unit second moment about y
[16] $I_{uxy} = \Sigma\ [1(x_m - x_c)(y_m - y_c) + 1 \cdot \Delta x \cdot \Delta y/12]$ [1](Table 9-2)
= about the centroid
= 0 mm³
Unit product moment
[17] $J_u = I_{ux} + I_{uy}$ [1](Table 9-1)
= 2266700 + 227560
= 2494200 mm³
Unit polar moment
[18] $T = F_y(e_x - x_c) - F_x(e_y - y_c) + M_z$
= -30 × 232.22 − 0 × 0 + 0e3
= -6.9667 kN·m
In-plane torque about the centroid
[19] $f_{r,max}$ = max over weld ends $\sqrt{(f_x^2 + f_y^2 + f_z^2)}$
= at (80, 0) mm: (-279.31, -257.13, 0) N/mm
= 379.64 N/mm
Maximum resultant force per unit length
[20] critical = critical point
= (80, 0) mm
= "(80, 0) mm"
Where f_r is greatest
[21] a = 0.707 * s
= 0.707 × 8 mm
= 5.656 mm
Throat
[22] $\tau = f_{r,max} / a$
= 379.64 N/mm / 5.656 mm
= 67.122 MPa
Maximum throat stress
[23] $s_{req} = f_{r,max} / (0.707 * f_w)$
= 379.64 N/mm / (0.707 × 220 MPa)
= 2.4408 mm
Leg size required
[24] $u = \tau / f_w$
= 67.122 MPa / 220 MPa
= 0.3051
Utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
$f_{r,max}$	379.6 N/mm	Maximum force per unit length
τ	67.1 MPa	Throat stress
s_{req}	2.4 mm	Leg required
L_w	360.0 mm	Weld length
x_c	17.8 mm	Centroid x
y_c	100.0 mm	Centroid y
I_{ux}	2266667 mm³	Unit second moment about x

SYMBOL	VALUE	DESCRIPTION
I_{uy}	227556 mm³	Unit second moment about y
I_{uxy}	0 mm³	Unit product moment
J_u	2494222 mm³	Unit polar moment
T	-6.97 kN·m	Torque about the centroid
critical	"(80, 0) mm"	Critical point
a	5.7 mm	Throat
u	0.31	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Throat stress against design strength	u = 0.31	u ≤ 1	0.31	[3](cl. 4.5.3.3)

7. Applicability

None.

8. Verification

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Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Blodgett, Design of Welded Structures.
3. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-8: Design of Joints. BS EN 1993-1-8:2005. BSI; 2005.